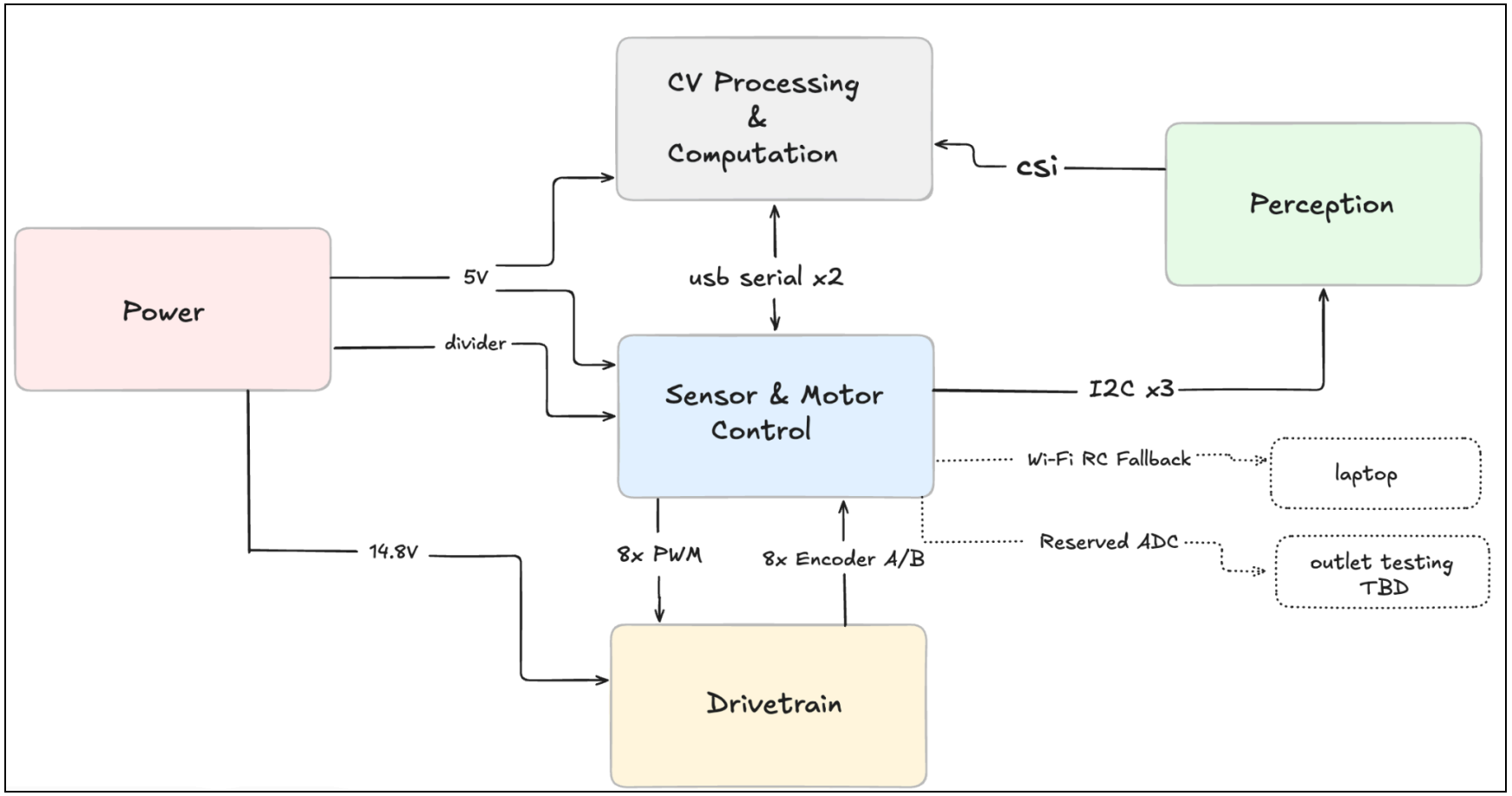
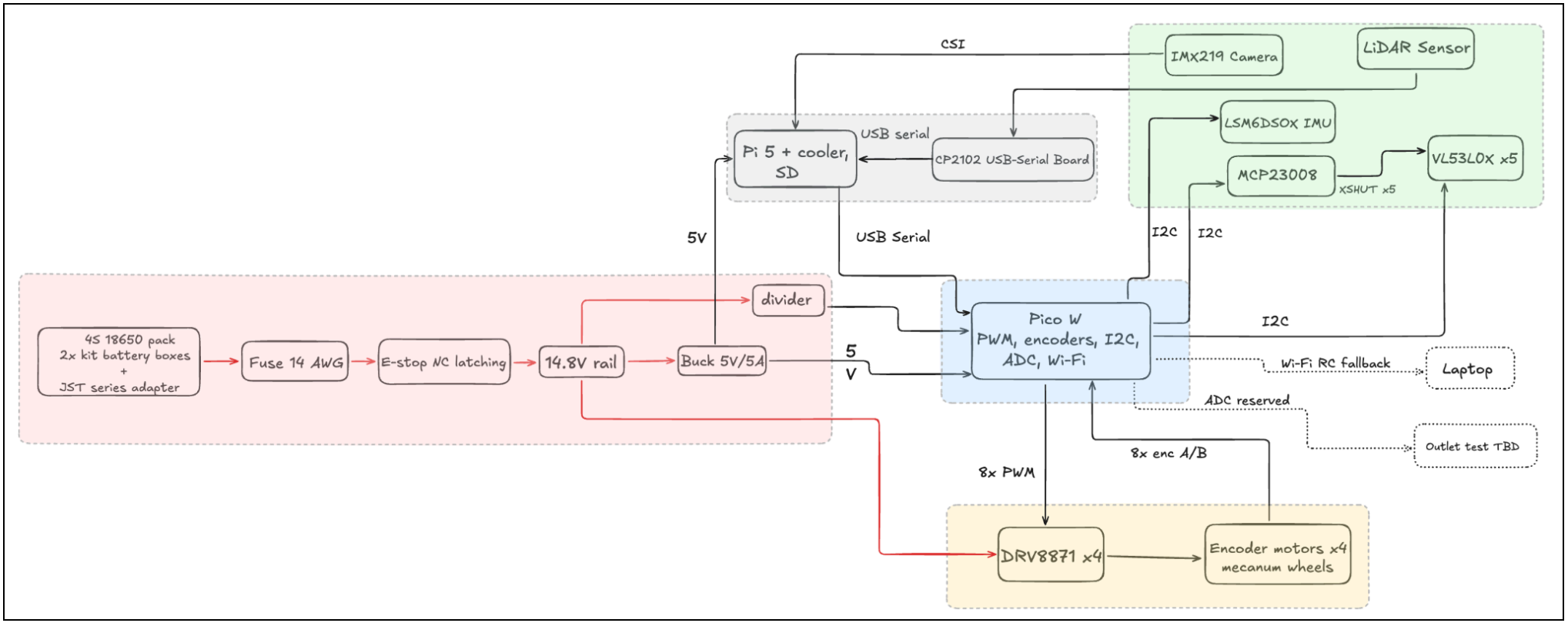


Top Level Diagram: Five functional blocks. Power conditions the 4S lithium-ion pack into the rails the system needs. Perception collects visual, ranging, and inertial data. CV Processing & Computation (Raspberry Pi 5) handles outlet detection, mapping, and navigation planning. Sensor & Motor Control (Pico W) handles the real-time layer, motor commands and sensor polling. Drivetrain converts commands into wheel motion and returns encoder feedback.



Component Level Diagram:
Each block expanded into its physical parts.

Wiring: red paths are the unregulated battery rail in 16 AWG (fuse holder leads are 14 AWG). Black paths are signal or low-current lines in 22 AWG.

Four 30Q cells in series form a 14.8V pack, routed through an inline fuse and a normally-closed latching E-stop that cuts all downstream power. The rail feeds the motor drivers directly and a 5V / 5A buck converter supplying the Pi 5, Pico W, lidar, and sensor breakouts. A resistor divider taps the rail ahead of the buck so the Pico can monitor pack voltage for under-voltage cutoff. The Pi 5 takes the IMX219 camera over CSI and the LD19 lidar over USB serial, and talks to the Pico over USB serial. The Pico runs one I2C bus carrying the IMU, an MCP23008, and the VL53L0X sensors, the expander drives their XSHUT lines so each can be readdressed at boot, since they share a default address. The Pico also generates 8 PWM channels into four DRV8871 drivers and reads 8 encoder channels back from the four mecanum-wheel motors.